

//////////////////////////////// MAIN MODULE////////////////////////////////

`timescale 1ns / 1ps

////////////////////////////////

// Company:

// Engineer:

//

// Create Date: 05.10.2026 14:45:26

// Design Name:

// Module Name: traffic_light_controller

// Project Name:

// Target Devices:

// Tool Versions:

// Description:

//

// Dependencies:

//

// Revision:

// Revision 0.01 - File Created

// Additional Comments:

//

////////////////////////////////

module traffic_light_controller (

input clk,

input reset,

input pedestrian,

output reg [2:0] highway,

output reg [2:0] side_road,

output reg ped_walk

```

);
//-----
// Light Encoding
//-----
parameter RED = 3'b100;
parameter YELLOW = 3'b010;
parameter GREEN = 3'b001;
//-----
// FSM States
//-----
parameter HW_GREEN = 2'd0;
parameter HW_YELLOW = 2'd1;
parameter SR_GREEN = 2'd2;
parameter PED_STATE = 2'd3;
reg [1:0] state;
reg [3:0] timer;
//-----
// FSM
//-----
always @(posedge clk or posedge reset)
begin
if(reset)
begin
state <= HW_GREEN;
timer <= 0;
highway <= GREEN;
side_road <= RED;
ped_walk <= 0;
end
end

```

```
else
begin
timer <= timer + 1;
case(state)
HW_GREEN:
begin
highway <= GREEN;
side_road <= RED;
ped_walk <= 0;
if(timer == 5)
begin
timer <= 0;
state <= HW_YELLOW;
end
end
HW_YELLOW:
begin
highway <= YELLOW;
side_road <= RED;
ped_walk <= 0;
if(timer == 2)
begin
timer <= 0;
if(pedestrian)
state <= PED_STATE;
else
state <= SR_GREEN;
end
end
```

```
SR_GREEN:
begin
highway <= RED;
side_road <= GREEN;
ped_walk <= 0;
if(timer == 5)
begin
timer <= 0;
state <= HW_GREEN;
end
end
PED_STATE:
begin
highway <= RED;
side_road <= RED;
ped_walk <= 1;
if(timer == 4)
begin
timer <= 0;
ped_walk <= 0;
state <= SR_GREEN;
end
end
endcase
end
end
endmodule
```

//////////////////////////////////TESTBENCH//////////////////////////////////

`timescale 1ns / 1ps

//////////////////////////////////

// Company:

// Engineer:

//

// Create Date: 05.10.2026 14:53:04

// Design Name:

// Module Name: traffic_light_controller_tb

// Project Name:

// Target Devices:

// Tool Versions:

// Description:

//

// Dependencies:

//

// Revision:

// Revision 0.01 - File Created

// Additional Comments:

//

//////////////////////////////////

module traffic_light_controller_tb;

reg clk;

reg reset;

reg pedestrian;

wire [2:0] highway;

wire [2:0] side_road;

```
wire ped_walk;

// DUT
traffic_light_controller DUT
(
    .clk(clk),
    .reset(reset),
    .pedestrian(pedestrian),
    .highway(highway),
    .side_road(side_road),
    .ped_walk(ped_walk)
);

// Clock
always #5 clk = ~clk;

// Test
initial
begin
    clk = 0;
    reset = 1;
    pedestrian = 0;
    #20;
    reset = 0;
    #80;
    pedestrian = 1;
    #10;
    pedestrian = 0;
    #150;
    $finish;
end

// Monitor
```

```
initial
begin
$monitor("Time=%0t | Ped=%b | Highway=%b | Side=%b | Walk=%b",
$time,
pedestrian,
highway,
side_road,
ped_walk);
end
endmodule
```